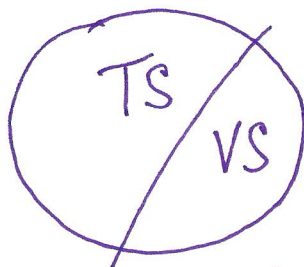


In Cross Validation
to obtain p^*

D.S.



(I 150)

This works if both TS & VS are "sufficiently large". Both must be statistically significant.

What to do if DS is not large enough to obtain two statistically significant sets ($\Rightarrow N$ is not "so big")

$\uparrow$
cardinality of the Data set.

Idea: if we had a number of data sets $\Rightarrow$ we could average on them. In principle, we would want to minimize

$$E \{ J_N(\hat{\theta}_p) \}$$

Where N could be "small".

Recall: two ideal Hypotheses:

① $S \approx M(\theta)$

[the model we are using can "perfectly fit" the system for some parameter vector θ]

② $\hat{\underline{\theta}}_p^* = \underline{\theta}_p$ [the optimal estimate (I.15) of the parameters corresponds to the ideal ones]

If (ideal) ① & ② hold true $\Rightarrow E \{ J_N(\hat{\underline{\theta}}_p^*) \} = E \left\{ \frac{1}{N} \sum_{k=1}^N e^2(k | \underline{\theta}_p) \right\} =$
 $= E \left\{ \frac{1}{N} \sum_{k=1}^N m^2(k) \right\} = \lambda^2$

Recall $m(k) \sim WN(0, \lambda^2)$

Of course, in general, & even if ① holds true,
 $\hat{\underline{\theta}}_p^* \neq \underline{\theta}_p$ because of the "lack of data" (limited dim of D.S.)
 & because the data set is noisy.

From Statistics:

i) if N is "big" $\Rightarrow E \{ J_N(\hat{\underline{\theta}}_p^*) \} \cong \lambda^2 \left(1 + \frac{p}{N} \right)$

where $p = \dim \underline{\theta}_p$

Reasonable: $N \rightarrow \infty \Rightarrow E \{ J_N(\hat{\underline{\theta}}_p^*) \} \rightarrow \lambda^2$
 p/N small $\Rightarrow$ far from overfitting

N = cardinality of Data Set.

λ^2 is unknown.

(I 152)

ii) An unbiased estimate of λ^2 can be obtained as

$$\lambda^2 \cong \frac{1}{N-p} \sum_{k=1}^N e^2(k|\hat{\underline{\theta}}_p^*)$$

to have an unbiased estimate.

$$\hat{\underline{\theta}}_p^* = \arg \min_{\hat{\underline{\theta}}_p} J_N(\hat{\underline{\theta}}_p)$$

Putting i) & ii) together $\Rightarrow$

$$\begin{aligned} E \{ J_N(\hat{\underline{\theta}}_p^*) \} &= \frac{1}{N-p} \frac{N+p}{N} \sum_{k=1}^N e^2(k|\hat{\underline{\theta}}_p^*) = \\ &= \frac{N+p}{N-p} \underbrace{\frac{1}{N} \sum_{k=1}^N e^2(k|\hat{\underline{\theta}}_p^*)}_{J_N(\hat{\underline{\theta}}_p^*)} = \underbrace{\frac{N+p}{N-p} J_N(\hat{\underline{\theta}}_p^*)}_{\text{F. P. E.}} \end{aligned}$$

F. P. E.
Final Prediction Error

This results suffers of the fact that N is indeed "not so big"

We can still consider these "approximations" (for a given D.S. of cardinality N) (I153)

$$\text{FPE}(p) = \frac{N+p}{N-p} J_N(\hat{\theta}_p^*)$$

$$\hat{\theta}_p^* = \arg \min_{\hat{\theta}_p} J_N(\hat{\theta}_p)$$

We can use FPE to optimize over p .

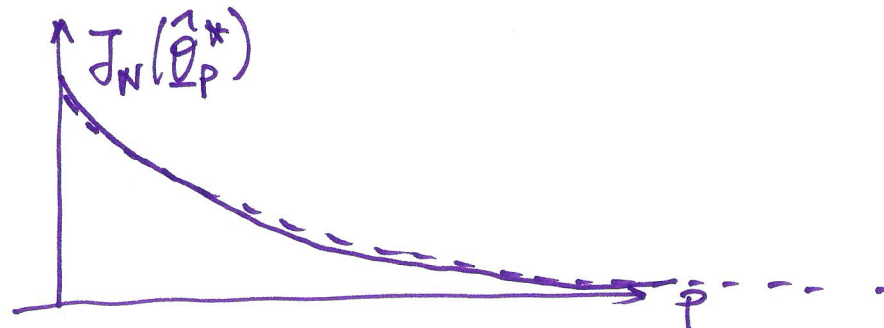
$$p^* = \arg \min_p \underbrace{\frac{N+p}{N-p}}_{g(p)} \underbrace{J_N(\hat{\theta}_p^*)}_{\frac{1}{N} \sum_{k=1}^N e^2(k | \hat{\theta}_p^*)}$$

Note that we here are using the full data set (N data)

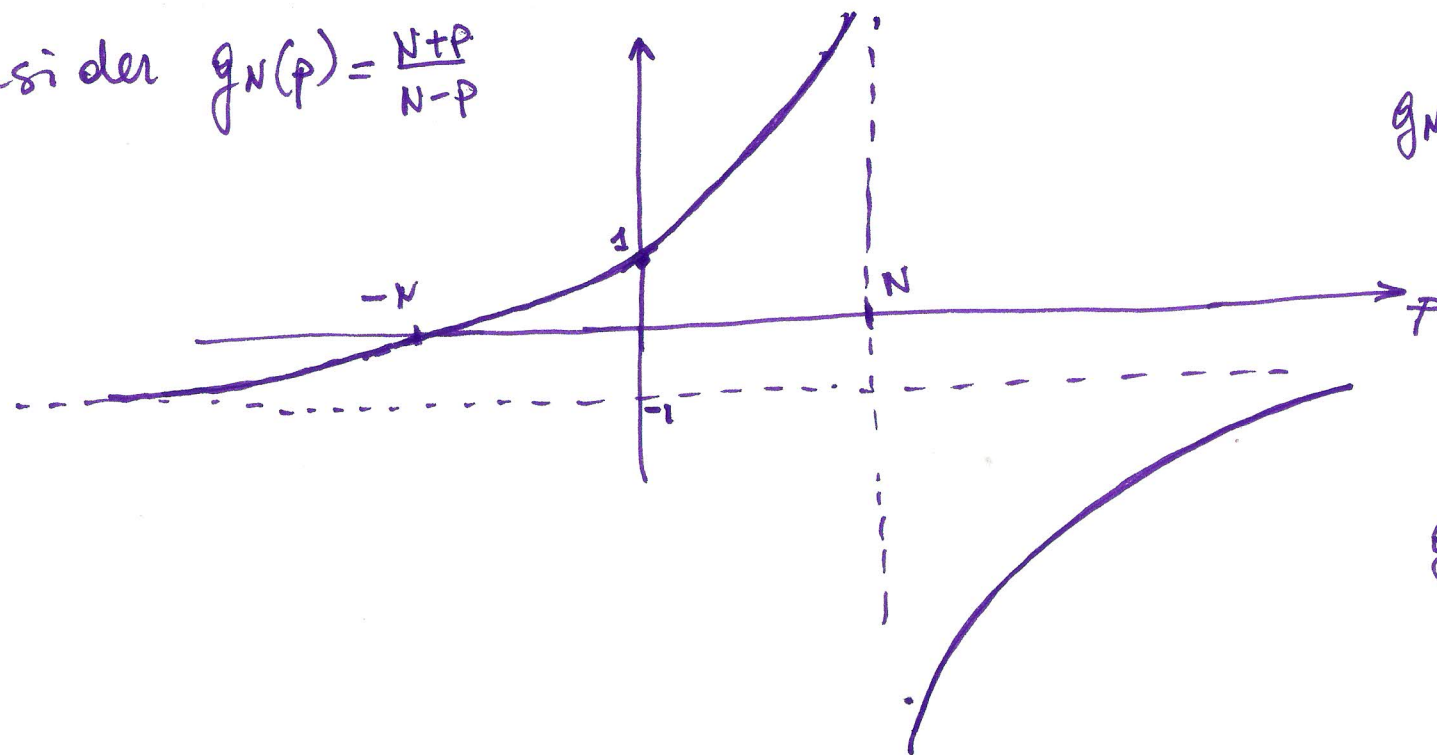
$$\hat{\theta}_p^* = \arg \min_{\hat{\theta}_p} \frac{1}{N} \sum_{k=1}^N e^2(k | \hat{\theta}_p)$$

$J_N(\hat{\theta}_p^*)$ is computed over the same dataset

Overfitting for large p



Consider $g_N(p) = \frac{N+p}{N-p}$



$$g_N(p) = \frac{1 + \frac{N}{p}}{-1 + \frac{N}{p}} \xrightarrow{p \rightarrow \infty} -1 \quad (I_{154})$$

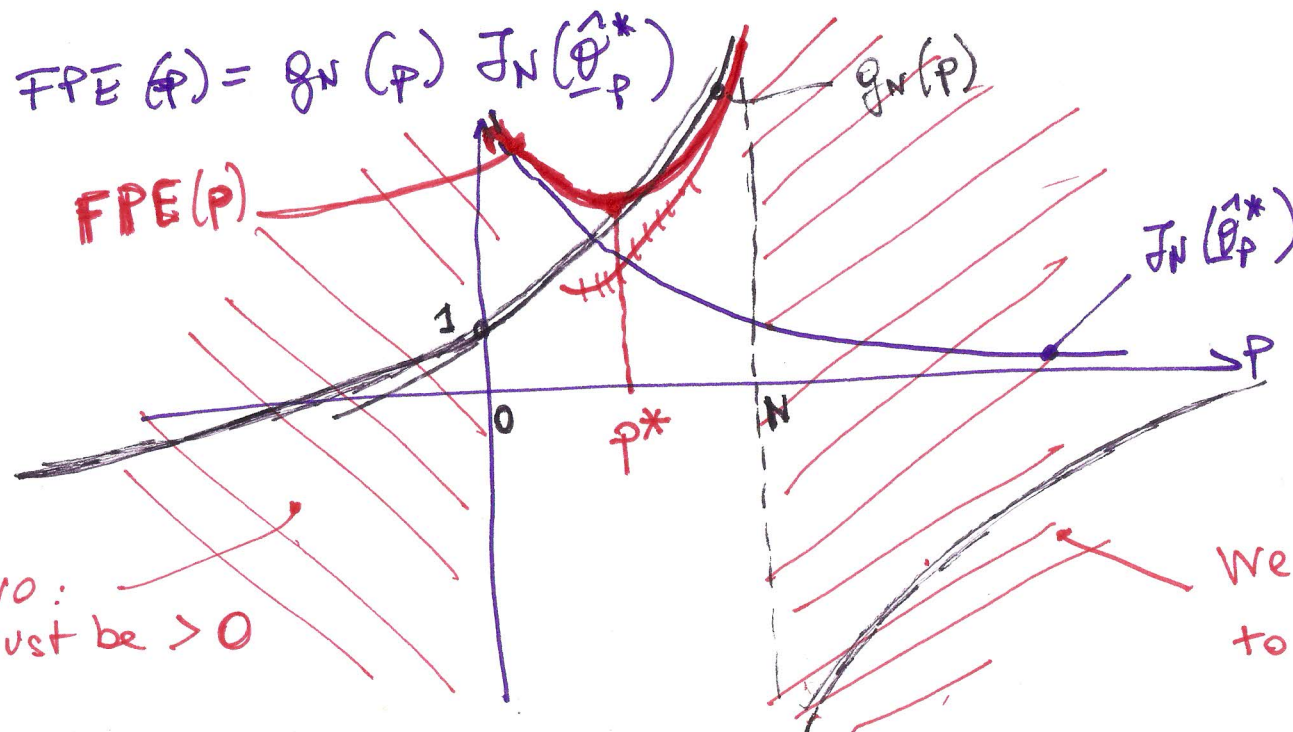
$$g_N(0) = 1$$

$p \rightarrow N$ vert. as.

$$g_N(p) = 0 \Leftrightarrow p = -N$$

$$FPE(p) = g_N(p) J_N(\hat{\theta}_p^*)$$

$FPE(p)$



NO:
 p must be > 0

We want $p < N$
to avoid OVERFITTING.

Multiplying $J_N(\hat{\theta}_p^*)$ by $g_N(p)$ let us avoid overfitting.

(I.155)

With only $J_N(\hat{\theta}_p) \Rightarrow$ minimum at $p=N$ Multiplying by $g_N(p) = \frac{N+p}{N-p}$ leads the choice to some "trade off" value p^*

$$0 < p^* < N$$

Other possibilities: - Akaike Information Criterion (AIC)
- Minimum Description Length (MDL)

$$AIC(p) = 2 \frac{p}{N} + \ln(J_N(\hat{\theta}_p^*))$$

$$MDL(p) = \frac{\ln N \cdot p}{N} + \ln(J_N(\hat{\theta}_p^*))$$

For $N \rightarrow \infty$ they all are "consistent".

For all ~~all~~ these criteria the idea is to use the full data set.

Ideal Hp: ① $S = \mathcal{M}(\underline{\theta})$

② $\hat{\underline{\theta}}^* = \underline{\theta}$

$$\Rightarrow e(k | \hat{\underline{\theta}}_p^*) = n(k)$$

Let consider B.B. models in pred. form.

Model

$$\hat{y}(k | \hat{\underline{\theta}}^*) = \left[1 - \frac{\hat{C}(z | \hat{\underline{\theta}}^*)}{\hat{D}(z | \hat{\underline{\theta}}^*)} \right] y(k) + \frac{\hat{B}(z | \hat{\underline{\theta}}^*) \hat{C}(z | \hat{\underline{\theta}}^*)}{\hat{A}(z | \hat{\underline{\theta}}^*) \hat{D}(z | \hat{\underline{\theta}}^*)} u(k)$$

Id. Syst:

$$y(k) = \left[1 - \frac{\hat{C}(z | \underline{\theta})}{\hat{D}(z | \underline{\theta})} \right] y(k) + \frac{\hat{B}(z | \underline{\theta}) \hat{C}(z | \underline{\theta})}{\hat{A}(z | \underline{\theta}) \hat{D}(z | \underline{\theta})} u(k) + n(k)$$

If ① & ② holds true ($\hat{\underline{\theta}}^* = \underline{\theta}$)

$$\Rightarrow y(k) - \hat{y}(k | \hat{\underline{\theta}}^*) = n(k)$$

We are assuming that $n(k) \sim WN(0, \lambda^2)$

Summing up: ① & ② $\Rightarrow e(k | \hat{\underline{\theta}}^*) \underset{n(k)}{\sim} WN(0, \lambda^2)$

If, after optimization of the parameters

$$e(k|\hat{\theta}^*) \quad \underline{\text{is NOT}} \quad WN(0, \lambda^2) \quad \text{for some } \lambda^2$$

(I 157)

① & ② $\Downarrow$ Do NOT jointly hold true.

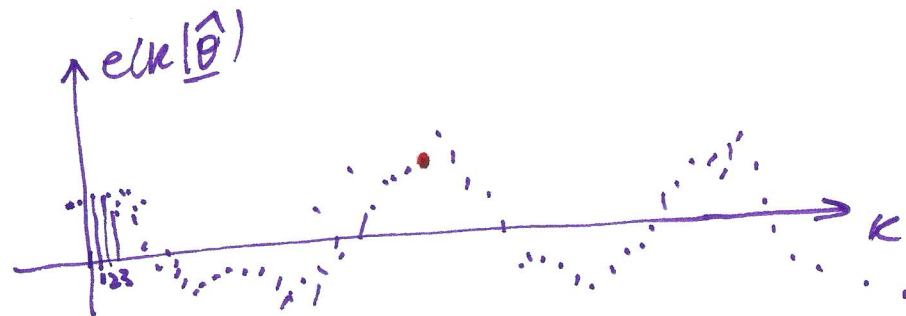
$\Rightarrow$ Whiteness Test for $e(k|\hat{\theta})$ for ~~checking~~ validating a model with param. $\hat{\theta}$.

For some param. vector $\hat{\theta}$

Compute $e(k|\hat{\theta}) = y(k) - \hat{y}(k|\hat{\theta})$

check if the sequence $e(k|\hat{\theta})$ is "white" with zero mean.

EX.



NOT WN

There is some dynamics in the errors.

$\Downarrow$
they are correlated over time.



This seems WN
but with
mean value $\neq 0$

(I.58)

This can be corrected by eliminating
the "bias" $\bar{e}$ in the model.

Possible Whiteness model $\Rightarrow$ ANDERSON TEST.

It starts choosing a confidence $0 < \epsilon < 1$

$\Rightarrow$ test results YES/NO "with confidence ϵ "